LAURENT "ENZO" NTIBARIKURE

Engineering | Energy & Information Systems | Authoring

@ ntilau@gmail.com   ntilau   ntilau   Eindhoven, Netherlands

SUMMARY

My experiences, with a bias on numerical methods, encompass electronic and mechatronic systems development, technical sales & support across several industries (Energy & Aerospace, Electronic Design Automation and Semiconductors).

Leveraging such a background, I am currently helping designers to develop more effectively towards system-level compliances.

EXPERIENCE

Sr. Engineer - Design Enablement

NXP Semiconductors

03/2022 – Present   Eindhoven, Netherlands

- Developed electromagnetic simulation flows for SoCs and antenna systems.
- Enhanced vendor solutions to meet design and cloud platform integration requirements.
- Developed problem resolution automation scripts and utilized parametric reduced order models.

Technical Account Manager

Ansys

02/2020 – 03/2022   Utrecht, Netherlands

- Drove customer satisfaction by providing multi-disciplinary technical support and managing global presales activities for the Ansys portfolio.
- Led teams to architect flows across disciplines, focusing on requirements and critical success factors.
- Advised global sales and account engineers on addressable initiatives and pipelines with preliminary requirements and solutions visions.

Sr. Applications Engineer

Ansys

01/2019 – 03/2020   Utrecht, Netherlands

- Executed presales activities for electronic systems across industries, supporting high frequency solutions and EMC flows across EMEA.
- Led technical alignment with customers on automotive electrification and connectivity projects, successfully delivering on consulting services and trainings.

Lead EIC Engineer - Testing

Baker Hughes

07/2016 – 12/2018   Florence, Italy

- Led the development of industrial, ATEX compliant, distributed data acquisition systems to streamline tests execution timelines from months to weeks.
- Resolved 5+ electromagnetic interference issues in NPI validations and factory acceptance tests, contributing to best practices.
- Received an innovation award for patent disclosure of the Structural Vibration Monitoring System and multiple rewards for innovative solutions and customer impact.

EIC Engineer - Testing

Baker Hughes

06/2013 – 06/2016   Florence, Italy

- Executed on electrical, instrumentation and controls design activities for GE Oil & Gas NTI/NPI (GT, CC, ST), focusing on uncertainty assessed special instrumentation (performance validation).

EDUCATION

Doctor of Philosophy - PhD, Electronics Engineering

Università degli Studi di Firenze

01/2011 – 02/2014   Florence, Italy

School of "RF, Microwaves and Electromagnetics". Computational Methods focus. With grant (Jan 2011 - Jun 2013).

Master's Degree, Electronics Engineering

Università degli Studi di Firenze

10/2007 – 04/2010   Florence, Italy

Erasmus Placement/LLP framework thesis semester at the Lehrstuhl für Theoretische Elektrotechnik, Universität des Saarlandes. With grant (Sep 2009 - Mar 2010).

Bachelor's degree, Electronics Engineering

Università degli Studi di Firenze

01/2003 – 04/2008   Florence, Italy

Thesis project at the Laboratorio di Tecnologia per i Beni Culturali, Università degli Studi di Firenze.

High School Diploma

École Autonome Belge à Bujumbura

09/1996 – 06/2002   Bujumbura, Burundi

AWARDS

EMEA TAM of the Year | Ansys (01/2021)
Award for customer and industry focus

Learn and adapt to win | BHGE (07/2018)
Patent impact award for the disclosure of "Module with sensors arrangement"

Customers determine our success | BHGE (03/2017)
Bronze award for the GE Aviation's Catalyst FETT advanced test instrumentation consulting

Customers determine our success | GE (10/2016)
Bronze award for the successful execution of the vibration monitoring system ZADCO customer structural vibrations acceptance tests

Learn and adapt to win | GE (03/2016)
Bronze award for the co-generation plant cost savings on the SNC1-2 steam turbine performance validation

- Contributed to rapid test executions through internal standardization and inventory optimization.

Startup Co-Founder

Università degli Studi di Firenze

06/2012 – 06/2013 Florence, Italy

- Initiated a spin-off, ElectroSoft, to provide efficient numerical codes for design problems.
- Led team promotion and value proposition pitches to business angels.
- Developed first opportunities in the defense industry.

PROJECTS

LM9000: 65 MW aeroderivative gas turbine

Baker Hughes, a GE Company

06/2018 - 12/2018 Florence, Italy

- Contributed to conceptual instrumentation design and test rigs erection, data acquisition architecture and control system design by technical and HAZOP reviews.

Bearings loading test rig

Baker Hughes, a GE Company

05/2018 - 12/2018 Florence, Italy

- Conceptual design of a novel bearings loading test rig aimed at validating bearings NTIs.
- Investigated on modeling methods of EM coupling between power (variable speed drive system) and signals cabling systems to predict potential interference issues.

Gas Turbine embedded SAW sensors

Baker Hughes, a GE Company

02/2018 - 12/2018 Florence, Italy

- Co-advised MEng thesis student with University of Florence to investigate into SAW sensors high frequency numerical modeling for accuracy control of temperature measurements.

Cross'd

01/2018 - 11/2018 Florence, Italy

- Social networking app PoC based on Bluetooth Low Energy ranging. Made with Apache Cordova and an AWS EC2 instance

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BLE Asset Tracking

07/2017 - 09/2017 Florence, Italy

- Sub 5$/Unit PoC of BLE tags (TI's CC2640) with firmware optimized for long-battery life, aimed at asset tracking and inventory management.

GE Aviation's Catalyst

Baker Hughes, a GE Company

06/2017 - 12/2018 Prague, Czech Republic

- Led special instrumentation design and global technical reviews of FETT with design and testing teams (Czech Rep., Poland, US, Canada)
- Contributed to >4 M$ cross-P&L project and customer satisfaction.

GE Power slip-ring digital telemetry

Baker Hughes, a GE Company

04/2017 - 05/2017 Florence, Italy

Learn and adapt to win | GE (10/2015)
Bronze award for the introduction and validation of an effective, cost-saving, structural vibrations digital integration algorithm aimed at accurately deriving speed & displacements measurements from ATEX-compliant accelerometers

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External focus | GE (06/2014)
Bronze award for the RCA activities on Cessao Onerosa, ADRE monitoring system EMI noise mitigation and customer assurance

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Giorgio Barzilai Prize | SIEm (09/2012)
Best paper award for young researchers, for the work "Model order reduction in Finite Element analysis of phased array antennas" presented at the XIX RiNEm

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Ph.D. studies grant | UniFi (12/2010)
Full tuition fees and allowance for research on Electromagnetic Compatibility modeling techniques in variable speed drive systems - with GE/Nuovo Pignone

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Dean's listing | UniFi (04/2010)
Commissione di Laurea Specialistica in Ingegneria Elettronica presieduta dal Prof. Piero Tortoli (Prot. 1051 n. class. III/9.1.2 del 24 Maggio 2010)

LLP/Erasmus Placement grant | UniFI (10/2009)
Thesis work semester mobility support at Universität des Saarlandes

CERTIFICATIONS

Professional Engineer, ID:ES2010277049000038
Università degli Studi di Firenze

COURSES

Command of the Message

Ansys

06/2020 Remote

JAWS: Just Another Way of Selling

Ansys

02/2020 Paris, France

HBM Dynamic Load Cells

GE Oil & Gas

05/2016 Bologna, Italy

GE Crotonville: Delivering Customer Impact

GE Oil & Gas

03/2016 Munich, Germany

GE Control Valves — Advanced

GE Oil & Gas

09/2015 Florence, Italy

- Ran Ansys HFSS simulations to optimize signal integrity over slip-ring connections (power supply and ethernet transmission lines)
- Achieved shrinking of the rings spacing leading to manufacturing costs reduction
- 4 weeks-FTE cross-P&L billing and customer satisfaction

Wet gas compressors at KLAB
GE Oil & Gas

02/2016 - 06/2016 Haugesund, Norway

- Designed and procured validation equipment for ATEX zone 1 test bench.
- Developed Ex-d antenna and low noise amplifier system for GPS synchronization on NI's PXI systems.
- Managed IEC 60079-11 simple apparatus compliance on load cells upon using extremely low voltage (1.5 V) Ex-ia Wheatstone bridge conditioners.

GE-6G PowerGen Module Structural Vibrations Monitoring System
GE Oil & Gas

11/2015 - 03/2016 Massa, Italy

- Developed ATEX-compliant distributed acquisition system for more than 150 vibration measurement points, with 40 kHz analog bandwidth, WiFi5 backhaul to storage and real-time monitoring, with GPS or daisy-chain clock distribution synchronization
- Hands-on design of 3D assemblies and BOM for timely delivery and cost optimization of certified equipment in less than 4 months (vs typ. 8 months).
- Introduced and consolidated with third party metrology a software based velocity and displacement real-time computation from accelerometers, achieving direct $150k savings, and subsequent lab processes OPEX benefits.
- Customer value delivered as onsite Turbomachinery Modules structural commissioning with patented solution

SNC1-2 steam turbine performance evaluation in 6.2 MW cogeneration plant
GE Oil & Gas

09/2015 - 04/2016 Florence, Italy

- Defined novel pressure hookups method at lower cost and better performance to achieve accuracy controlled inter-stage enthalpy measurements according to ASME PTC 19.11.
- Programmed a CAN to Modbus TCP embedded module to enable static torque measurements on a MagCanica telemetry.

NovaLT16 gas turbine, bearings and cooling component tests
GE Oil & Gas

04/2014 - 10/2015 Florence, Italy

- Design and validation of smartbox (field datalogger) for analog signals.
- Defined EMC-aware measurement loops design to achieve noise immunity and enhanced measurement accuracies, rigorously applying EN 61000 guidelines (CENELEC25).

Power density centrifugal compressor
GE Oil & Gas

03/2014 - 05/2016 Florence, Italy

- Led the commissioning Fieldbus Foundation use in testing lab for more than 240 pressure transducers (0.1% uncertainty) upon identifying the proper Firmware configuration which enabled for massive and accurate pressure measurements.
- Managed capacitive vibration probes ATEX-compliance.
- Developed electrostatic numerical model to predict capacitance variation in various surroundings and target shape conditions.
- Contributed to DM37/08 compliance of the test rig contract specification, including power and signaling cabling selection and proper installation.

CE Marking Overview
GE Oil & Gas

11/2014 Florence, Italy

Safety Integrity Levels
GE Oil &Gas

10/2014 Florence, Italy

Electrification
GE Oil & Gas

10/2014 Florence, Italy

GE EHS Orientation
GE Oil & Gas

10/2013 Florence, Italy

ATEX Intermediate
GE Oil & Gas

09/2014 Florence, Italy

Total Quality Lean Six Sigma Green Belt
GE Oil & Gas

06/2014 Florence, Italy

PidXp and P&ID
GE Oil & Gas

11/2013 Florence, Italy

Intellectual Property
GE Oil & Gas

11/2013 Florence, Italy

GE Oil & Gas Product Familiarization
GE Oil & Gas

07/2013 Florence, Italy

SKILLS

Engineering | Simulations | Testing | Management | Sales | Strategy | Leadership | Customer Satisfaction | Finite Element Analysis | RF | Electronics | Semiconductors | Scripting | Matlab | C++ | C | Python

LANGUAGES

Dutch	1/5
English	4/5
French	5/5
German	1/5
Italian	5/5
Kirundi	5/5

GE Aviation's GE9X High Pressure Compressor

GE Oil & Gas

01/2014 - 04/2014 Massa, Italy

- High density data acquisition system design for GE Aviation's compressor test in Massa.
- Performed mitigation on EMC noise due to 130 kW variable speed ventilation system

BCL306 and BCL317 centrifugal compressors rotordynamics

GE Oil & Gas

06/2013 - 03/2014 Florence, Italy

- Performed an Ex-p execution of stability's magnetic exciter, contributed to the resolution of customer's opened nonconformity on rotordynamic vibrations upon conducting an electromagnetic interference on Bently Nevada's 3500 RCA.

Finite Element Software (FES)

Università degli Studi di Firenze

06/2011 - 02/2014 Florence, Italy

- FES is an open source framework for the development of "ad-hoc" simulation software aimed at a rapid implementation of novel numerical methods based on the Galerkin method. Core formulations verified reusing meshes from commercial packages (Comsol, HFSS)

Phased array antennas model order reduction (MEng thesis)

Universität des Saarlandes

10/2009 - 02/2010 Saarbrücken, Germany

- Developed an intrusive model order reduction in finite element analysis of antenna arrays for fast and accurate beamforming
- Implemented Matlab code for a 3000x matrix reduction and 300x speedup in antenna pattern computation

RS-232 modem in ISM 868 MHz band (BEng thesis)

University of Florence

01/2007 - 04/2007 Florence, Italy

- Developed a wireless connectivity for a battery-powered avalanche rescue ground penetrating radar, optimizing firmware for fast full-duplex/half-duplex conversion timing.
- Executed the design and assembly of an ISM 868 MHz serial modem PCB prototype with off-the-shelf MCU and Radio (sub-GHz RFFE).
- Achieved successful wireless system resulting in improved breath signal detection quality.

RECOMMENDATIONS

Mariano Morales

Senior Director, Global Technical Account Management @ Ansys (A Synopsys company)

23/05/2022 Madrid, Spain

> "During his tenure with Ansys, Laurent performed consistently above expectations. He showed a tremendous ability to master new technology domains out of his comfort zone, to team up with colleagues from different functional groups and demonstrated an extreme customer focus that made him succesful. He is always eager to learn and evolve as a technical leader."

Tiziano Fiore

Head of Global Product Management @ ABB

09/09/2020 Helsinki, Finland

Spanish ●●●○○

PUBLICATIONS

Patents

- L. Ntibarikure, "Module with sensors arrangement," US20180180464A1, 2018. [Online]. Available: https://patents.google.com/patent/US20180180464A1/en

Books

- L. Ntibarikure, *Contributions to the Art of Finite Elements in Electromagnetics*. Florence, Italy, 2014. [Online]. Available: https://hdl.handle.net/2158/843133

Journal Articles

- L. Ntibarikure, "Multiphysics simulations for 5g rfics and socs," *Bits&Chips*, 2019.
- F. Barone, A. Signorini, L. Ntibarikure, T. Fiore, F. Di Pasquale, and C. J. Oton, "Fiber-optic liquid level sensing by temperature profiling with an fbg array," *Sensors*, vol. 18, no. 8, 2018. [Online]. Available: https://www.mdpi.com/1424-8220/18/8/2422
- L. Ntibarikure, G. Pelosi, and S. S. and, "Harmonic balance domain decomposition finite elements for nonlinear passive microwave devices analysis," *Electromagnetics*, vol. 34, no. 3-4, pp. 239–252, 2014. DOI: 10.1080/02726343.2014.877756
- L. Ntibarikure, G. Pelosi, and S. Selleri, "Efficient harmonic balance analysis of waveguide devices with nonlinear dielectrics," *IEEE Microwave and Wireless Components Letters*, vol. 22, no. 5, pp. 221–223, 2012. DOI: 10.1109/LMWC.2012.2192420

Conference Proceedings

- L. Ntibarikure, G. Pelosi, and S. Selleri, "Assessment of the performances of gmres(r) using a domain decomposition approach as a preconditioner," in *Proceedings, XX Riunione Nazionale di Elettromagnetismo (XX RiNEm)*, Padua, Italy, 2014. [Online]. Available: https://inspirehep.net/literature/1403858
- L. Ntibarikure, "Model order reduction in finite elements analysis of phased array antennas," in *Proceedings, XIX Riunione Nazionale di Elettromagnetismo (XIX RiNEm)*, S. d. E. SiEm, Ed., Roma, Italy, 2012.
- L. Ntibarikure, G. Pelosi, and S. Selleri, "Harmonic balance finite element analysis of third order intermodulation products in ferrite devices," in *Proceedings, XIX Riunione Nazionale di Elettromagnetismo (XIX RiNEm)*, S. d. E. SiEm, Ed., Roma, Italy, 2012.

“Laurent is a brilliant engineer with entrepreneurial mindset. He approaches technical problems critically and with a deep knowledge of physics that allows him to handle any engineering challenge. Laurent is a great manager and a strong comunicator always willing to help others and share ideas and solutions with anyone in the team. I have learnt a lot from him, sitting side by side, working together on challenges in turbomachinery validation projects. I greatly recommend Laurent for business development and executive roles.”